questions on a weekly basis; this benchmark only offers a very small subset (48 questions) on which to evaluate LLMs against human performance.

A major issue with all three dynamic benchmarks is domain-specific relevancy. For example, ForecastBench includes questions on the ELO scores of professional chess players, and the Metaculus AI Forecasting Benchmarking Series includes questions on podcast rankings. To the best of our knowledge, these questions are irrelevant from a policy or economic perspective.

3 Methodology

3.1 Notation and Preliminaries

In this work, we consider judgmental binary forecasting, wherein an LLM is presented with some binary question q. Toward answering the question q in an informed manner, we allow the LLM to gather some evidence, $\mathcal{E}$, before attempting its answer. For example, we might query an online news database of articles for articles related to q.

Using the gathered evidence, the LLM makes a prediction p, which denotes the probability of the outcome under scrutiny occurring. If the LLM is absolutely sure the event in question will come to pass, it should set $p = 1.0$. Likewise, if the LLM is sure the event will not happen, it should set $p = 0.0$. Intermediate values such as $p = 0.3$ should be used if the LLM is uncertain about the final outcome, but nonetheless has some evidence which shades its beliefs in a particular direction.

This process of mulling over evidence and assigning a probability can be denoted as

$$\pi : (q, \mathcal{E}) \to p$$

Of course, we have hidden a great many difficulties with the explanation above. Principal among them, we supposed the existence of an evidence gathering pipeline $\mathcal{E}$, tasked with collecting information that is useful for informing the LLM's predictions. As good human researchers know, evidence gathering is often a highly iterative process: some article contains a key piece of information, which sparks further research, cascading in dozens of queries to find the information required to fully understand the underlying situation. Human forecasters may also work collaboratively, exchanging information and engaging in debate to arrive at a final forecast.

More generally, in designing an LLM forecaster, we face a vast array of design decisions, many of which mirror those faced by expert human forecasters. These choices include the choice of retrieval pipeline, the method of reasoning over evidence, and the manner in which to evaluate performance, among others. Below we describe the design of the AIA Forecaster, which takes inspiration from the human forecasting process.

3.2 The AIA Forecaster

Architecture. The AIA Forecaster is a multi-agent forecasting system, in which several agents perform an adaptive search procedure to retrieve relevant information, reason over the retrieved content, and finally synthesize this reasoning to produce a single probability. The system then reconciles these independent forecasts and applies a statistical correction to produce a final answer.

A full visualization of our forecasting pipeline can be found in Figure 1. In the pipeline, we see that M individual agents are spawned. Each agent carries out independent research and produces an initial forecast probability. Those individual forecasts are passed to a supervisor agent, which identifies disagreements among the M agents and is given a final chance to query our news database with any clarifying questions. After clarifying, the reconciliation pipeline outputs an aggregated forecast, which is then passed through a Platt scaling step to produce the final probability.

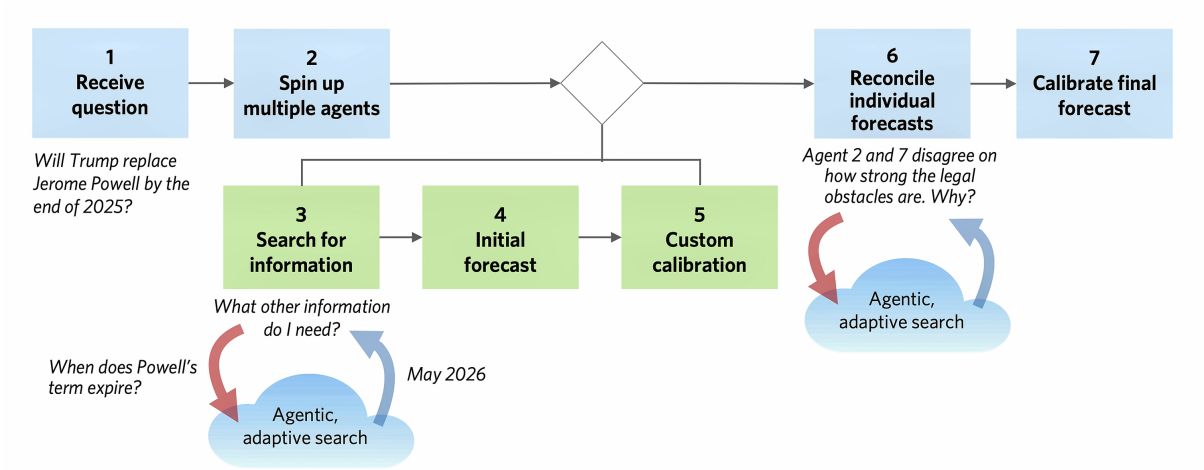

Figure 1: The architecture of the AIA Forecaster

We now describe the individual components of the AIA Forecaster.

Search. Each forecasting agent is granted access to an external search provider, which it can query as needed. As described above, a key feature of this search procedure is that it is *agentic* and *adaptive*: each agent is granted full discretion to determine whether and how to query the search provider, and can iteratively improve the quality of its evidence by conditioning each query on prior results. Each forecasting agent operates independently, and their paths can diverge significantly over the course of a single run. As we discuss below, this can be regarded as both a bug and a feature. We can denote this multi-stage evidence gathering pipeline in mathematical notation as

$$\pi : q \to \mathcal{E}_1 \to \mathcal{E}_2 \to \cdots \to \mathcal{E}_n \to p$$

While agentic search produces far better results than the non-agentic search considered in prior work (Halawi et al., 2024; Karger et al., 2024), this change is insufficient to achieve superforecasting levels. There are multiple reasons for this, all of which can be distilled into the fact that individual forecast runs are unstable, and this instability effectively destroys the fidelity of aggregated benchmark results. The instability itself comes from many sources: missing a key piece of information, over-anchoring to an incorrect base rate, and the inherent randomness of LLMs leading to the occasional outlier opinion.

Reconciliation. One naive solution for addressing run-over-run instability is to simply execute forecasts many times from scratch and to ensemble the results. We show that taking a naive mean over N forecasts is a very strong baseline, and that many advanced ensembling methods fail to outperform this comparatively simple strategy.

In particular, suppose we try to beat the mean by asking an LLM to carefully read all the individual forecasts, evaluate their relative merits, and then produce a final estimate (as proposed in Halawi et al. (2024)). When we do this, we see somewhat surprisingly that the resulting aggregated prediction is substantially less accurate than a simple average of predictions. By inspecting the reasoning traces, we find that LLMs are prone to overemphasizing outlier opinions. This failure exacerbates the instability inherent to LLM forecasting, putting us in a difficult position.

However, it turns out that a simple reframing of this problem is enough to turn our fortunes around. Rather than asking a supervisor LLM to aggregate all forecasts directly, we instead instruct a supervisor agent to examine all forecasts' reasoning traces $\mathcal{R}_i$ and identify sources of ambiguity or disagreement. This supervisor is allowed to output N search queries, with the hopes that these will help reconcile the differences between competing forecasts. Useful queries at this stage often involve looking up a base rate, or fact-checking assertions made by individual forecasting agents.

Thus, if we denote information gathering agent i as π_i, our fully realized evidence gathering pipeline can be written as

$$\pi_i : q \to \mathcal{E}_1 \to \mathcal{E}_2 \to \cdots \to \mathcal{E}_n \to (\mathcal{R}_i, p_i)$$
$$\text{Supervisor} : (\mathcal{R}_1, \mathcal{R}_2, \ldots, \mathcal{R}_M) \to \mathcal{E}_{\text{supervisor}} \to p_{\text{final}}$$

Hedging. Finally, we have one more piece of drama to address. It turns out that LLMs have an annoying bias, likely stemming from post-training via reinforcement learning from human feedback (RLHF) (Kaufmann et al., 2023), which makes them hedge toward 0.5 when making predictions. Forecasts for events where the outcome is somewhat certain (e.g., a true probability of 0.85) are often attenuated to a more entropic forecast (e.g., 0.6). Additionally, even for questions where the outcome is certain, LLMs will often predict 0.95 rather than 1.0, justifying their response with reasoning like "unexpected things might happen, and we need to be safe." We document some examples in Appendix G. These